CHEMISTRY LAB PREPARATION LIST FOR JC2 H2 CHEMISTRY

2018 PRELIM EXAM

APPARATUS NEEDED (PER STUDENT)

S/N	Apparatus	Quantity	Location	Remarks
1	10 cm^3 measuring cylinder	2	Student's bench	Reuse
2	50 cm^3 measuring cylinder	1	Student's bench	Reuse
3	25 cm^3 measuring cylinder	1	Student's bench	Reuse
4	100 cm^3 measuring cylinder	1	Student's bench	Reuse
5	stopwatch	1	Student's bench	Reuse
6	Thermometer (-10°C to +110°C	1	Student's bench	Reuse
7	Glass rod	1	Student's bench	Reuse
8	Wash bottle of distilled water	1	Student's bench	Reuse
9	Pen suitable for labelling glassware	1	Student's bench	Reuse
10	Heat proof mat	1	Student's bench	Reuse
11	Filter Funnel	1	Student's bench	Reuse
12	Test tube rack	1	Student's bench	Reuse
13	Test tube holder	1	Student's bench	Reuse
14	Bunsen burner	1	Student's bench	Reuse
15	Tripod with gauze	1	Student's bench	Reuse
16	Lighter	1	Student's bench	Reuse
17	Spatula	1	Student's bench	Reuse
18	100 cm^3 conical flask	1	Student's bench	Reuse
19	Stand and **bigger clamp**	1	Student's bench	Reuse
20	plastic tub	1	Student's bench	trough for gas collection (minimum capacity 1 dm_3)
21	Paper towels		Student's bench	Must change to new one per shift
22	Test tubes	6	Student's bench	Must change to new one per shift
23	Boiling tubes	2	Student's bench	Must change to new one per shift
24	Filter Paper	2-3	Student's bench	Must change to new one per shift
25	Dropping Pipette	5	Student's bench	Must change to new one per shift

* access to a balance weighing

CHEMICALS NEEDED (PER STUDENT)

S/N	Chemical	Concentration	Quantity	Location	Remarks
1	Hydrochloric acid (**FA 1**)	2.0 mol dm^{-3}	70 cm^3		Dilute 170 cm^3 of concentrated (35-37%; approximately 11 mol dm^{-3}) hydrochloric acid to 1 dm^3
2	Magnesium (ribbon) (**FA 2**)	about 4 cm length	3 pieces		4 cm Magnesium ribbon supplied in a stoppered container labelled **FA 2** .
3	Sodium thiosulfate (**FA 3)**	0.10 mol dm^{-3}	200 cm^3		Dissolve 24.82g of $Na_2S_2O_3.5H_2O$ in each dm^3 of solution.
4	Hydrochloric acid (**FA 4)**	1.0 mol dm^{-3}	200 cm^3		Dilute 500 cm^3 of 2.0 mol dm^{-3} HCl to 1 dm^3 with distilled water.
5	manganese(IV) oxide, MnO_2 (**FA 5**)	-	About 6 g		In a stoppered weighing bottle labelled **FA 5**;
6	sulfuric acid (**FA 6**)	0.5 mol dm^{-3}	40 cm^3		Labelled **FA 6** Cautiously pour 27.5 cm^3 of concentrated (98%) sulfuric acid into 500 cm^3 of distilled water with continuous stirring. Make the solution up to 1 dm^3 with distilled water. **Care:** concentrated H_2SO_4 is very corrosive **[MH]**
7	Solid sodium ethanedioate (**FA 7)**	-	about 2 g		In a stoppered bottle labelled **FA 7**
8	manganese(II) sulfate, $MnSO_4$ **FA 8**)	About 0.5 mol dm^{-3}	3 cm^3		dissolving, in deionised water, about 84.5 g of $MnSO_4.H_2O$ in each dm^3 of solution. This solution must be stored in a a **stoppered** vial labelled **FA 8**
9	cyclohexene, C_6H_{10}, (**FA 9**)	-	about 1 cm^3		in a **stoppered** vial labelled **FA 9**;
10	bromine water, $Br_2(aq)$ (**FA 10**)	-	about 2 cm^3		in a **stoppered** vial labelled **FA 10**.

STANDARD BENCH REAGENTS

Hazard	Label	Identity	Notes (hazards given in this column are for the raw materials)
[MH]	Dilute hydrochloric acid	2.0 mol dm^{-3} HCl	Dilute 170 cm^3 of concentrated (35 – 37%; approximately 11 mol dm^{-3}) hydrochloric acid to 1 dm^{-3}
[C]	Dilute nitric acid	2.0 mol dm^{-3} HNO_3	Dilute 128 cm^3 of concentrated (70% w/v) nitric acid to 1 dm^3
[MH]	Dilute sulfuric acid	1.0 mol dm^{-3} H_2SO_4	Cautiously pour 55 cm^3 of concentrated (98%) sulfuric acid into 500 cm^3 of distilled water with continuous stirring. Make the solution up to 1 dm^3 with distilled water. **Care:** concentrated H_2SO_4 is very corrosive
[C][MH][N]	Aqueous ammonia	2.0 mol dm^{-3} NH_3	Dilute 112 cm^3 of concentrated (35%) ammonia to 1 dm^3
[C]	Aqueous sodium hydroxide	2.0 mol dm^{-3} NaOH	Dissolve 80.0 g of NaOH in each dm^3 of solution. **Care:** the process of solution is exothermic and any concentrated solution is very corrosive.
[MH]	Aqueous barium nitrate	0.1 mol dm^{-3} barium nitrate	Dissolve 26.1 g of $Ba(NO_3)_2$ in each dm^3 of solution.
[N]	Aqueous silver nitrate	0.05 mol dm^{-3} silver nitrate	Dissolve 8.5g of $AgNO_3$ in each dm^3 of solution.
[MH]	Limewater	Saturated aqueous calcium hydroxide, $Ca(OH)_2$	Prepare fresh limewater by leaving distilled water to stand over solid calcium hydroxide for several days, shaking occasionally. Decant or filter the solution.
	Aqueous potassium manganate (VII)	0.02 mol dm^{-3} potassium manganate (VII)	Dissolve 3.16g of $KMnO_4$ in each dm^3 of solution.
	Aqueous Iodine		Dissolve the 2.5g of solid in water containing sufficient potassium iodide to dissolve the iodine, then make up the solution to 1 dm^3 with distilled water.

The following materials and apparatus should be available **FOR EACH STUDENT**.

Red and blue litmus papers, plain filter strips for use with acidified manganate(VII), aluminium foil for testing nitrate/nitrite, wooden splints and the apparatus normally used in the Centre for use with limewater testing for carbon dioxide.